
A Capable, Fully Tensor-Decomposable Vision-Language-Action Policy

Exact rational conversion, protocol-aligned control, and weight-derived causal
structure

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Abstract

Tensor decomposition factors a large table of interactions into a small set of low-rank components that can be ranked and inspected. Applying it directly to a conventional robot policy is difficult because softmax, ordinary activations, and normalization interrupt the required algebraic form. We introduce χ -VLA, a compact vision-language-action policy whose learned map uses only bilinear, linear, and rational operations. Its checkpoint therefore defines an explicit rational tensor program, making interactions among image, instruction, state, and action coordinates accessible from the weights. A mechanical audit finds no hidden incompatible operation, and local reconstructions agree near the float32 precision floor. On all ten LIBERO-Object tasks, the 20.1M-parameter policy reaches $92.8\% \pm 2.1\%$ with a ViT encoder and $93.7\% \pm 0.8\%$ with a convolutional encoder under the published three-seed protocol. A same-skeleton conventional twin differs by only 0.006% in parameters. Complete same-hardware replications find a 4.4-point rational deficit on A30 and 5.8 points on A6000. Compilation reduces the latency ratio from $3.49\text{--}5.33\times$ in eager mode to $1.73\text{--}1.74\times$ across the two GPU classes. We derive an exact weight-based decomposition of softmax-free attention and reconstruct every attention module in three checkpoints with worst relative error 2.56×10^{-7} . All 36 primary block-head tests favor its rank-128 subspace over equal-rank random controls, with median fidelity ratio 2.44. At the policy’s visual bond, retaining 64 of 384 causally ranked dimensions preserves 18/20 closed-loop successes, compared with 19/20 for the full representation and 0/20 for a random subspace. A 300-pair instruction intervention across three checkpoints shows consistent language-driven motion but incomplete scene-prior override. The evidence establishes competitive specialist capability and exact layerwise access while exposing nonzero seed-0 capability and systems costs. It does not establish broad generalization or selective coefficient-level repair.

1 Introduction

Modern robot policies can work well while remaining difficult to inspect. A checkpoint contains millions of learned numbers, but it does not directly reveal which combinations of image regions, instruction tokens, and robot state produce an action. Analysts therefore usually collect activations on a dataset and fit a separate probe. That can identify correlations, but it leaves open whether the same structure is present in the weights and whether it causes behavior.

Tensor decomposition offers a different analysis surface. A tensor is a multidimensional table, and a decomposition rewrites that table as a sum of simpler low-rank factors, much as a matrix factorization

finds dominant directions in a matrix. If a network is built from additions, multiplications, and ratios of polynomials, its weights define an explicit table of input interactions. Decomposing that table can rank the combinations the model is constructed to use. By *fully tensor-decomposable*, we mean that every learned block preserves this algebraic form. The aim is not merely compression. It is a direct path from parameters to candidate mechanisms that can be tested by intervention [3, 4].

Robot control makes this goal useful and difficult. Visual shortcuts can produce high benchmark success while ignoring the instruction, and small action errors compound in closed loop. Standard softmax, nonlinear activations, and square-root normalization break the explicit polynomial form. Continuous actions also cannot rely on the external categorical sampling used by a language model. A practical design must recover these functions without losing precise control.

We study the narrower but testable question:

Can a specialist VLA retain competitive closed-loop capability when every deployed learned operation is polynomial or rational, and does the resulting structure provide a useful weight-based analysis surface?

The stronger thesis has three parts: decomposability should have small capability cost, the weights should expose stable behaviorally causal interactions, and editing those interactions should selectively change behavior without retraining. This paper establishes strong capability and exact layerwise structure, then states clearly where causal diagnosis and editing remain incomplete.

Our contributions are:

1. We construct a complete VLA from tensor-compatible primitives. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We evaluate two fully rational 20.1M-parameter policies under a protocol aligned with published LIBERO-Object results. The strongest single architecture reaches $93.7\% \pm 0.8\%$ over three seeds, and a three-checkpoint ensemble reaches 96.2%. A same-skeleton conventional twin differs by only 0.006% in parameters. Two complete same-hardware replications place the seed-0 capability cost between 4.4 and 5.8 points.
3. We mechanically audit the real trained policy and explicitly reconstruct deployed rational and bilinear branches from their weights.
4. We extend exact weight-derived decomposition through individual attention layers with a reduced-Gram calculation that avoids materializing the full high-order tensor.
5. We show a causal closed-loop consequence. A selected 64-dimensional visual subspace preserves almost all tested behavior, while a random equal-width subspace fails. A separate three-checkpoint instruction intervention distinguishes reliable trajectory response from incomplete task-level rerouting.

Table 1 separates the evidence already established from the tests still needed for the stronger thesis. This distinction is important because algebraic representability does not by itself imply tractable analysis, and a causally important activation subspace is not yet a coefficient-level editing interface.

Conjunct	Current evidence	Decisive missing test
Small capability cost	The rational policy reaches $92.8\% \pm 2.1\%$ over three seeds. A matched seed-0 twin shows a 4.4 to 5.8 point cost across two GPU classes.	Complete conventional seeds 1 and 2 and partial-conversion controls, then localize the task-specific gap.
Weight-only diagnosis	Exact layerwise attention decomposition consistently beats fixed random subspaces.	Recover a concrete behavioral condition from weights before viewing its test rollouts.
Predictive surgery	A selected activation subspace causally preserves behavior.	Edit specified coefficients, predict the direction, and measure target effect and collateral damage without retraining.

Table 1: **Status of the three-part thesis.** Strong capability and exact layerwise structure are established. Hardware-controlled cost and coefficient surgery remain open.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. The empirical question is whether the algebraic policy makes them more direct, stable, and selectively editable than matched post-hoc methods.

2 A rational tensor robot policy

2.1 Tensor-compatible primitives

Each primitive below replaces a familiar transformer component while keeping an explicit coefficient description. Linear maps contribute first-order terms. Multiplying learned linear projections creates higher-order interactions whose coefficients remain functions of the weights, rather than an unstructured nonlinear black box.

Homogeneous coordinate. We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

Bilinear feed-forward block. For $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D [(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

The inner sum is an order-three table indexed by output coordinate and two input coordinates. The three matrices give its CP factorization, the tensor analogue of a low-rank matrix factorization, without materializing every table entry.

Softmax-free bilinear attention. Each head forms two query-key systems and one value stream:

$$Y = C [M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top) / s] V. \quad (2)$$

Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key, value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations. Unlike softmax attention, every multiplicative interaction can be expanded into coefficients determined by the trained weights.

Rational per-instance normalization. RMS normalization contains an input-dependent square root. We instead deploy a fixed rational approximation $r(z) = P(z)/Q(z)$ to $1/\sqrt{z} + \epsilon$:

$$\text{RNorm}(x) = x r \left(\frac{1}{d} \sum_{j=1}^d x_j^2 \right). \quad (3)$$

Here *rational* means a ratio of two polynomials. Projective coordinates carry each activation as a numerator-denominator pair (N, D) representing N/D . Linear, bilinear, residual, and normalization operations update this pair. The model therefore keeps input-specific rescaling while remaining exactly representable as a rational tensor network.

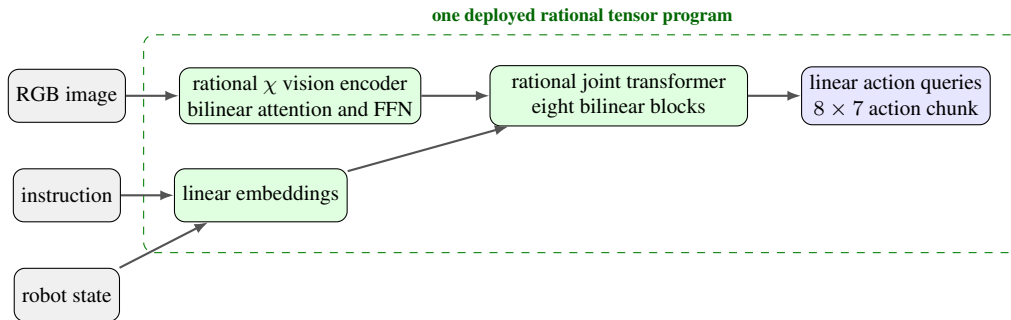


Figure 1: **Deployed χ -VLA map.** The strongest checkpoints use one vision stream and a linear continuous-action head. Every learned component inside the dashed boundary is linear, bilinear, or rational.

98 2.2 Architecture and training

99 The policy follows the high-level VLA sequence of visual encoding, language and state embedding,
 100 joint processing, and action decoding [1]. It uses a 64×64 agent-view image, an eight-dimensional
 101 robot state, a 26-token task vocabulary, and eight action-query tokens. The joint backbone has width
 102 384, eight blocks, and twelve heads. The action head maps each action-query state directly to seven
 103 continuous action dimensions. The complete model has 20,137,352 learned parameters.

104 Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay,
 105 bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera
 106 rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks
 107 that verify every requested state was loaded.

108 2.3 Mechanical certificate on the trained checkpoint

109 Architecture definitions can differ from deployed code. We therefore audit the real 189/200 check-
 110 point at runtime and reconstruct representative operations from saved weights.

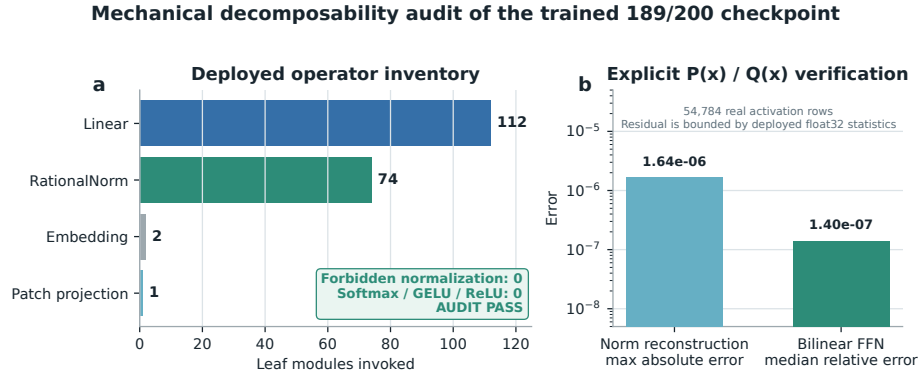


Figure 2: **Mechanical audit of the trained rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.

111 The normalization reconstruction has maximum absolute error 1.64×10^{-6} . A bilinear FFN re-
 112 construction has median relative error 1.40×10^{-7} . These residuals are bounded by the deployed
 113 module’s float32 statistic. The audit certifies operator membership and local reconstruction. It does
 114 not materialize one compact polynomial for the full eight-layer transformer.

115 3 Protocol-aligned closed-loop capability

116 3.1 Evaluation protocol

117 We evaluate all ten LIBERO-Object tasks with:

- 118 • 50 canonical trials per task
- 119 • a 280-step cap
- 120 • three independently trained seeds per architecture
- 121 • 500 rollouts per seed and 1,500 rollouts per architecture

122 Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not
 123 rerun in our codebase, so differences may still include implementation and training choices.

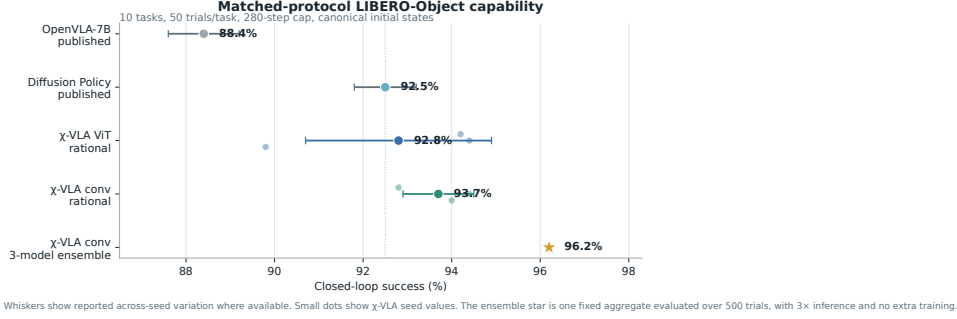


Figure 3: **Protocol-aligned LIBERO-Object capability.** Small dots are χ -VLA training seeds. The ensemble star averages three convolutional checkpoints and uses three forward passes per action chunk. External references are reported published values.

3.2 Results

Method	Parameters	Training seeds	Success
χ -VLA, rational ViT	20.1M	3	$92.8\% \pm 2.1\%$
Same-skeleton conventional ViT	20.1M	1	89.8%
χ -VLA, rational conv	20.1M	3	$93.7\% \pm 0.8\%$
χ -VLA, conv ensemble	$3 \times 20.1\text{M}$	$3 \rightarrow 1$	96.2%
OpenVLA-7B [1]	7B	3	$88.4\% \pm 0.8\%$
Diffusion Policy [1]	not matched	3	$92.5\% \pm 0.7\%$

Table 2: **Closed-loop success under aligned evaluation settings.** The two external rows are historical references rather than current state-of-the-art claims.

The ViT seeds score 89.8%, 94.4%, and 94.2%. The convolutional seeds score 94.0%, 94.4%, and 92.8%. The convolutional model is modestly higher and more consistent in this three-seed comparison.

The same-skeleton conventional control replaces bilinear attention, bilinear FFNs, and rational normalization with softmax attention, SwiGLU, and RMSNorm. It keeps the observations, token layout, demonstrations, optimizer, updates, seed, action decoder, canonical initial states, and trial budget fixed. On A30, the conventional policy succeeds in 448/500 trials and the rational policy in 426/500. On A6000, they reach 451/500 and 422/500. The rational model has 20,137,352 parameters and the conventional model has 20,138,632, a difference of 0.006%. The rational deficit is therefore 4.4 and 5.8 points. A stratified trial bootstrap gives intervals of $[-8.0, -0.8]$ and $[-9.4, -2.2]$ points, respectively. These intervals quantify the fixed-checkpoint canonical states, not across-seed uncertainty. Task 3 accounts for 21 of the lost successes on both platforms. On one A30, eager batch-one latency is 38.05 ms for the rational model and 7.14 ms for the conventional control. Compilation reduces these to 1.52 and 0.88 ms, a $1.74\times$ gap. At batch 128, rational training is $3.12\times$ slower and uses $1.63\times$ the peak allocated memory. A6000 profiling independently reduces the eager $3.49\times$ gap to $1.73\times$ after compilation.

Elementwise averaging of the three convolutional action predictions reaches 96.2%. This is 2.3 points above the mean of its members and 1.2 points above the strongest member. The gain is largest on tomato sauce, butter, and ketchup. The ensemble requires three complete forward passes and is not one 20.1M-parameter policy.

3.3 What this result does and does not establish

The result shows that the rational architecture supports high closed-loop specialist capability and closes the first same-seed capability-cost comparison. One conventional seed is not enough for a three-seed noninferiority claim, and the eager latency gap is material. Additional conventional seeds, partial-conversion controls, and optimized execution are still required before claiming that convertibility is broadly or computationally free.

4 Exact weight-derived structure through attention

4.1 Exact layerwise construction

Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives a high-order coefficient table. Each entry says how strongly one combination of query, key, and value coordinates contributes to an output. It is *weight-derived* because no examples or fitted probe are needed to construct it. On a tiny four-token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to 2.00×10^{-15} . The internal core identity agrees to 1.24×10^{-14} .

The deployed attention also normalizes each query and key with Equation 3. Each normalization multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled outside the bilinear dot products. The polynomial attention core is unchanged, while explicit numerator and denominator factors carry the normalization. The rational extension agrees to 3.61×10^{-16} . We then reconstruct all four vision and eight joint attention modules in each of three trained checkpoints, covering 384 heads in total. The worst relative ℓ_2 error is 2.56×10^{-7} and the worst absolute difference is 2.23×10^{-5} . This residual comes from the deployed implementation’s intentional float32 statistic, while the reconstruction itself uses the learned coefficients.

Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15} values. We instead contract the CP factors into an $O(d^2)$ reduced Gram. This smaller matrix plays the role of a covariance matrix: its leading eigenvectors rank the directions that carry the most coefficient energy without constructing the full table. At toy multi-head scale, it matches brute-force materialization to 2.13×10^{-14} .

4.2 Trained policy result

For the subspace-ranking experiment, we apply the exact weight-derived calculation to all twelve heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are sampled once, then fixed and reused for all evaluation examples. The weights choose the subspace. Real cached activations are used only afterward to measure how faithfully that subspace preserves the layer’s computation.

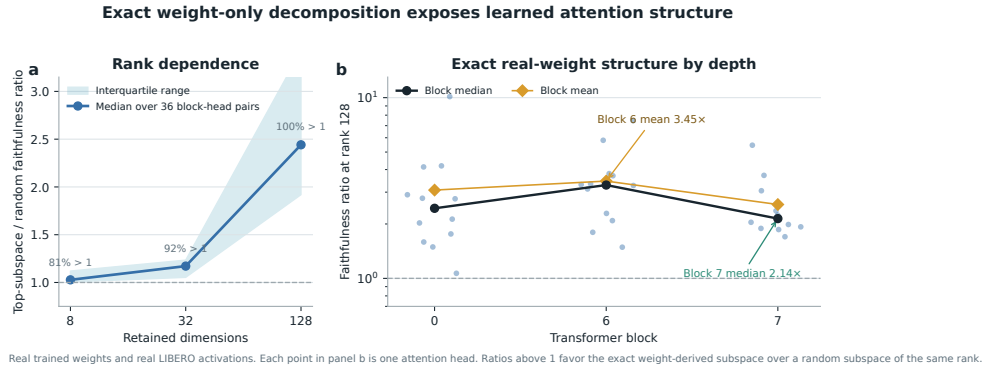


Figure 4: **Exact weight-derived attention structure.** Ratios above one favor the exact weight-derived subspace over a random equal-rank subspace. Each point in the right panel is one trained attention head.

At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks 0, 6, and 7.

This extends exact weight-only construction through individual attention layers. It is not an exact end-to-end decomposition of the complete policy. Composition across all eight blocks still faces rapid degree and representation growth.

5 Causal policy structure and a language shortcut

We next test whether the identified structure affects closed-loop behavior and whether high task success reflects robust use of the language input.

5.1 Causal visual subspace

At the visual bond before the joint backbone, we rank directions separately for translation, rotation, and gripper actions. A subspace is a smaller coordinate system inside the model’s 384-dimensional visual representation. Retaining a subspace means zeroing every direction outside it, then running the actual policy. We compare the selected directions with random equal-width controls in offline reconstruction and closed-loop intervention. This ranking uses held-out activations and downstream sensitivity, so it tests a causal bottleneck but is not a weight-only discovery result.

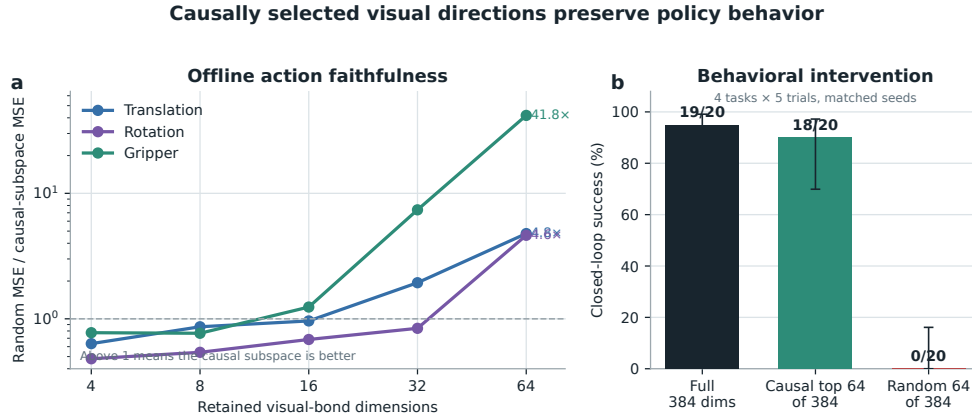


Figure 5: **Causally selected visual directions preserve policy behavior.** The left panel reports random MSE divided by selected-subspace MSE. The right panel intervenes on the real policy over four tasks and five paired trials per task.

With 64 of 384 dimensions retained, the selected subspace reduces action reconstruction error relative to random by $4.8\times$ for translation, $4.6\times$ for rotation, and $41.8\times$ for the gripper. In closed loop, the full representation succeeds in 19/20 trials, the selected subspace in 18/20, and a random subspace in 0/20.

This result establishes a causal low-dimensional substrate, but it is still limited to one checkpoint and twenty rollouts. The current ranking at this full-policy bond uses held-out examples and downstream sensitivity. The exact layerwise attention method in Section 4 must still be connected to the same full-suite closed-loop intervention.

5.2 Counterfactual instruction test

High success does not guarantee robust language use. A one-step screen is weak: replacing the instruction with one naming another visible object shifts the predicted reach direction toward that object in only 15.0% of 80 canonical states. We therefore run paired 80-step rollouts from 100 matched canonical states for each of three checkpoints. The instruction change increases relative end-effector preference for the renamed object by 0.175, 0.168, and 0.189 m. All three trial-bootstrap 95% intervals exclude zero. The trajectories shift toward the renamed object in 84%, 86%, and 88% of pairs, but end closer to it in only 33%, 28%, and 28%. Appendix Figure 7 reports the replication. The seed-0 conventional twin responds even more strongly, with a 0.239 m shift, 92% directional response, and a 33% endpoint rate. Instruction responsiveness is therefore not unique to the algebraic architecture. This is behavioral grounding evidence, not counterfactual task success, because the benchmark predicate still names the original object and the horizon covers the reach phase.

The likely shortcut is structural in the dataset. Each scene template is paired with an almost fixed target, so scene identity can substitute for object-name grounding. Decorrelated scripted demonstrations

improve the counterfactual metric, but all three fine-tuning attempts severely reduce ordinary closed-loop success. The best repaired dataset raises capability only to 21.5%, far below the healthy range.

This mixed result narrows the VLA claim. Language changes the trajectory reproducibly, but scene priors often prevent complete rerouting. The model is therefore a capable language-conditioned specialist, not yet a robust instruction-grounded policy. LIBERO-CF independently documents the same broad failure class in existing VLAs [10]. Our prospective novelty is therefore not the shortcut alone. It is the harder test of whether exact weight-derived terms can localize and selectively repair that shortcut.

6 Decisive remaining tests

1. Complete the controlled conversion study. Complete seeds 1 and 2, add partial-conversion controls, and retain the predeclared three-point noninferiority margin. Seed 0 misses that margin on both tested GPU classes. Report FLOPs, training throughput, memory, eager latency, and optimized latency separately. The current evidence supports strong rational capability and a modest nonzero seed-0 cost.

2. Direct coefficient surgery. Our first preregistered four-task screen directly edits query, key, and value columns selected by exact attention Grams. Retention gives a favorable 10-point gap over random controls but misses the 15-point gate, while removal has the wrong sign. This is a no-go for the current edit mapping, as detailed in Appendix A.7. The next test must resolve redundant paths or gauge sensitivity, compare post-hoc controls across three checkpoints, and report targeted effect, ordinary success, edit norm, and collateral damage.

3. Concrete diagnosis and selective repair. Develop two capstones. Gripper timing is the lower-risk target because phase transitions and premature closure have explicit simulator variables. Scene identity versus instruction binding is the higher-upside target because the current policy demonstrably fails the counterfactual test. In both cases, discover the algebraic term from weights, attach a limited semantic label, freeze the candidate, predict the behavioral consequence, and test on blind paired rollouts. A compelling repair should improve the targeted failure by 10 to 20 points, lose no more than 2 to 3 points of ordinary success, and preserve its direction across three seeds.

4. Broader capability and grounding. Evaluate the conventional and rational finalists on LIBERO-Goal and LIBERO-Long, followed by a counterfactual grounding benchmark such as LIBERO-CF [10]. Object-only success is insufficient because standard LIBERO can reward scene-to-action memorization [9]. Test the stability of recovered mechanisms under equivalent reparameterizations and across trained seeds.

5. RL only after a paired gate. The repaired-reward AWR pilot improves 80.7% to 82.7%, but it has one seed and fifteen trials per task. A justified follow-up needs three paired seeds, a matched-compute behavior-cloning continuation control, full protocol evaluation, and regression gates. Generic PPO on saturated LIBERO-Object is lower priority than the four tests above.

7 Related work

VLA capability. OpenVLA established an open pretrained VLA and standardized LIBERO comparisons [1]. OpenVLA-OFT showed that continuous action chunks and simple regression can substantially improve adaptation and control efficiency [2]. These systems use large pretrained backbones. χ -VLA instead studies a compact from-scratch specialist under an algebraic restriction.

Tensor and polynomial networks. Polynomial networks and tensor networks use explicit multilinear structure for representation and learning [12–14]. Bilinear MLP work showed that weight eigendecomposition can expose low-rank features and circuits [3]. χ -nets introduced ODT for compositional bilinear networks [4]. We extend the setting to a continuous closed-loop policy and give an exact reduced construction for individual softmax-free attention layers.

Causal mechanisms and policy analysis. Bilinear IMPALA policies have combined competitive ProcGen control with weight-derived eigenfilters, followed by activation probing and control [5]. This is the closest architectural precedent for weight-based policy analysis. Recent work steers pretrained VLAs through causally identified activation directions [6]. Sparse feature circuits identify and edit causal feature graphs in conventional language models [7], while nonlinear causal reductions recover behavioral patterns in reinforcement-learning policies from rollout perturbations [8]. These results make a universal separation from conventional policies untenable. Our intended distinction is empirical: exact weight-derived discovery and direct coefficient edits in a competitive continuous vision-language policy, evaluated for stability and collateral damage against matched post-hoc baselines.

VLA robustness. Recent evaluations show that standard LIBERO success can coexist with weak robustness and language use [9–11]. Our instruction intervention reaches the same conclusion from an independently trained compact specialist and motivates a decomposition-guided diagnosis.

8 Limitations and conclusion

χ -VLA demonstrates that a complete rational tensor policy can reach strong specialist closed-loop capability. The real checkpoint passes a mechanical operator audit. Exact weight-derived analysis now crosses individual attention layers, and a compact causal visual subspace preserves almost all tested behavior.

Five limits determine the next version of the paper. First, the same-skeleton control has one completed seed, a 4.4 to 5.8 point capability gap, and a residual compiled-runtime tax. Second, the benchmark is one narrow suite and counterfactual instructions do not fully reroute the policy. Third, exact decomposition is layerwise rather than end to end. Fourth, the closed-loop causal intervention uses one checkpoint and twenty trials. Fifth, activation projection succeeds but the first direct coefficient-edit screen fails its selectivity gate, so no behaviorally validated weight edit has yet been established.

The defensible conclusion is therefore narrow. Polynomial and rational restrictions do not prevent a compact policy from reaching competitive LIBERO-Object capability. The matched seed-0 control isolates a modest nonzero cost that now requires multi-seed explanation. The restrictions also provide a precise weight-derived analysis surface. Establishing multi-seed conversion cost, broad language-grounded control, semantic diagnosis from weights, and a decomposition-enabled repair requires the controlled experiments in Section 6. If all three conjuncts in Table 1 hold on the same trained policies, the result is stronger than any individual capability, decomposition, or intervention claim.

Reproducibility. Training and evaluation use fixed task lists, explicit camera and action conventions, canonical-state manifests, checkpointed moving-average weights, and per-run result files. The final artifact will release code, checkpoint hashes, exact attention certificates, and per-episode evaluation logs. Preflight checks fail if canonical initial states cannot be loaded.

References

- [1] M. J. Kim et al. *OpenVLA: An Open-Source Vision-Language-Action Model*. CoRL, 2024.
- [2] M. J. Kim, C. Finn, and P. Liang. *Fine-Tuning Vision-Language-Action Models: Optimizing Speed and Success*. 2025.
- [3] A. Rigg, T. Doods, M. T. Pearce, J. Oramas, and L. Sharkey. *Bilinear MLPs Enable Weight-Based Mechanistic Interpretability*. ICLR, 2025.
- [4] T. Doods, W. Gauderis, G. A. Wiggins, and J. Oramas. *Compositionality Unlocks Deep Interpretable Models*. 2025.
- [5] F. Oozer, Y. Eriskin, A. Rigg, J. Oramas, and L. Sharkey. *Bilinear Convolution Decomposition for Causal RL Interpretability*. 2024.
- [6] B. Häon, K. Stocking, I. Chuang, and C. Tomlin. *Mechanistic Interpretability for Steering Vision-Language-Action Models*. 2025.

- [7] S. Marks et al. *Sparse Feature Circuits: Discovering and Editing Interpretable Causal Graphs in Language Models*. ICLR, 2025.
- [8] M. Kekić et al. *Learning Nonlinear Causal Reductions to Explain Reinforcement Learning Policies*. ICLR, 2026.
- [9] X. Zhou et al. *LIBERO-PRO: Towards Robust and Fair Evaluation of Vision-Language-Action Models Beyond Memorization*. 2025.
- [10] Y. Fang et al. *When Vision Overrides Language: Evaluating and Mitigating Counterfactual Failures in VLAs*. 2026.
- [11] C. Kim et al. *LIBERO-Para: A Diagnostic Benchmark and Metrics for Paraphrase Robustness in VLA Models*. 2026.
- [12] T. Kolda and B. Bader. *Tensor Decompositions and Applications*. SIAM Review, 2009.
- [13] E. Stoudenmire and D. Schwab. *Supervised Learning with Tensor Networks*. NeurIPS, 2016.
- [14] G. Chrysos et al. *Deep Polynomial Neural Networks*. CVPR and TPAMI, 2020 to 2021.

A Supporting results and decisions

A.1 Original 94.5% capability result

Before the final protocol-aligned benchmark, one rational ViT checkpoint succeeded in 189/200 trials under a 600-step horizon and 20 trials per task. This result established that the rational implementation could support all ten tasks. It should not be compared directly with the protocol-aligned external references because it uses one training seed, fewer trials, and a longer horizon.

A.2 Matched architecture capability and systems cost

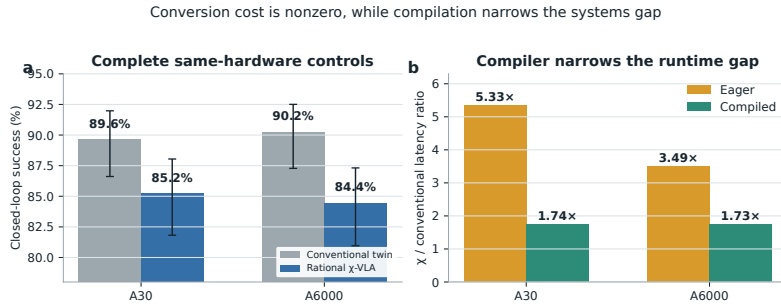


Figure 6: Conversion cost is modest but nonzero, and compilation narrows the systems gap. Complete same-hardware evaluations give 89.6% versus 85.2% on A30 and 90.2% versus 84.4% on A6000, conventional then rational. Parameters differ by 0.006%. Compilation narrows the batch-one latency ratio to 1.73–1.74 $\times$ on both GPUs.

332 A.3 Three-checkpoint counterfactual grounding

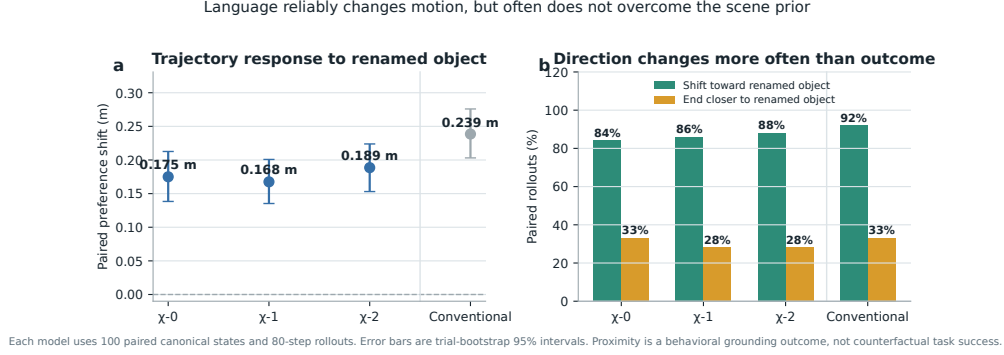


Figure 7: **Renaming the target changes motion reliably but does not fully override the scene prior.** Each model is evaluated on 100 paired canonical states for 80 control steps. The left panel reports the change in end-effector preference for the newly named object, with 95% bootstrap intervals. The right panel separates directional response from the stricter condition of ending closer to that object. These are proximity tests, not counterfactual task-success measurements. The conventional control shows that ordinary language response is not unique to tensor decomposability.

333 A.4 Per-task ensemble behavior

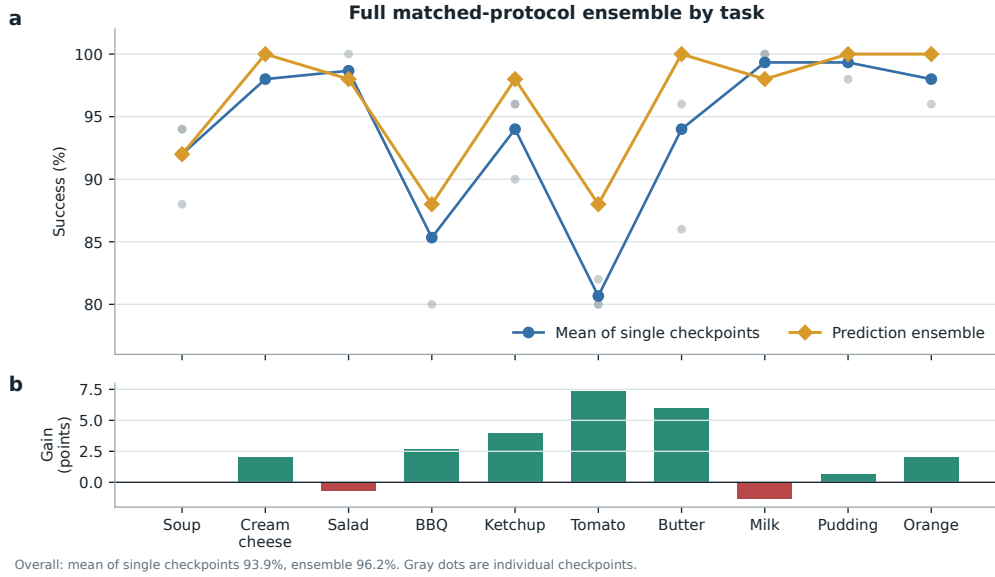


Figure 8: **Per-task effect of prediction averaging.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 20.1M-parameter policy.

334 A.5 Product-routing multimodal head

335 For an action-query state h , the product-routing head represents

$$a(h, b) = c_0(h) + \sum_{g=1}^G (2b_g - 1)c_g(h), \quad b \in \{0, 1\}^G. \quad (4)$$

336 The center, offsets, and gates are bilinear CP modules. A final external sign choice selects one of
 337 2^G modes. The head folds to numerical precision, but its five-seed closed-loop mean is $44\% \pm 30\%$
 338 and its range overlaps the linear baseline. The strongest rational policy therefore uses the linear head.
 339 Product routing remains a construction result until it wins a controlled genuinely multimodal task.

340 A.6 Development interventions

341 Focused DAgger on BBQ and tomato improves combined success from 82.5% to 96.0% using seven
 342 corrective episodes and 1,131 frames. Applying the same update thinly across all ten tasks reduces
 343 overall success from 92.0% to 72.0%. Three original AWR runs decline by 8.0, 4.6, and 2.6 points.
 344 A reward audit shows that accumulated step cost makes every successful trajectory have negative
 345 return. After removing that term and repairing the terminal boundary, one pilot improves by two
 346 points. These are development diagnostics, not validated paper contributions.

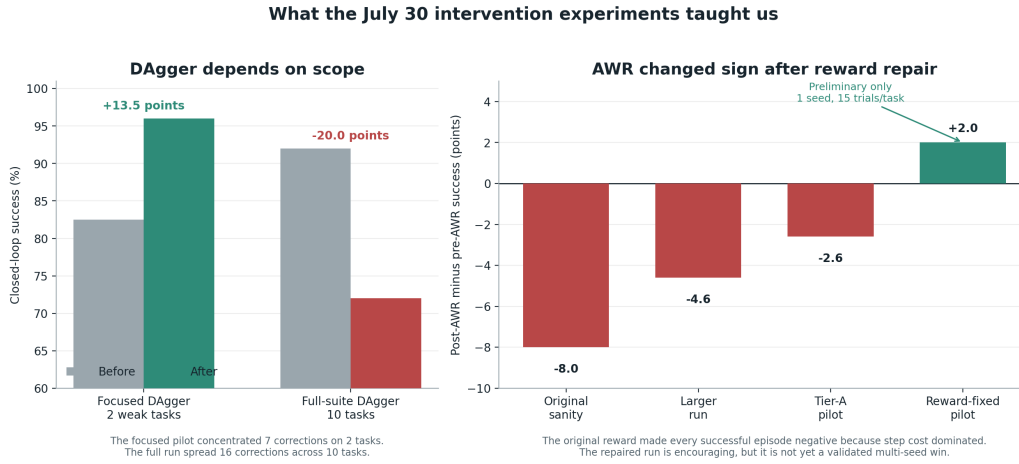


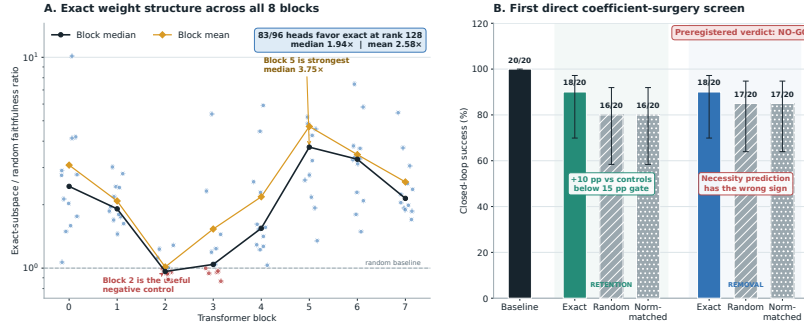
Figure 9: **Development interventions reveal scope and reward sensitivity.** Focused DAgger helps two weak tasks, while spreading a small correction budget across ten tasks causes regression. Three AWR variants also regress before a reward audit and repair. The positive reward-repaired AWR result is a one-seed pilot with 15 trials per task, so it is not evidence of a validated improvement.

347 A.7 Expanded attention screen and first direct edit

348 The corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128,
 349 83/96 block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean
 350 is 2.58, and blocks 2 and 3 provide useful weak or negative controls. At ranks 8 and 32, only 54/96
 351 and 60/96 pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at
 352 the wider tested rank.

353 The first direct weight-edit screen uses four tasks and five paired canonical episodes per task. Exact-
 354 subspace retention preserves 18/20 successes compared with 16/20 for both random controls. This
 355 10-point advantage misses the preregistered 15-point gate. Exact removal also preserves 18/20,
 356 compared with 17/20 for both random controls, which is the wrong direction for the necessity
 357 prediction. The screen is a no-go for the current mapping from Gram eigenspaces to query, key, and
 358 value column edits.

Weights expose reproducible structure, but the first direct edit is not yet selective



A. 96 block-head pairs, fixed random-control banks, rank 128. Exact directions come from weights; cached activations only score layer faithfulness. B. 4 tasks $\times$ 5 paired canonical episodes. Direct QK/V weight edits, no hooks, gradients, retraining, or rollout fitting. Error bars are 95% Wilson intervals.

Figure 10: **Expanded weight-structure and direct-surgery results.** Left: all 96 block-head pairs at rank 128, with fixed random-control banks. Exact directions come from weights, while cached activations only score layer faithfulness. Right: the first direct query-key-value column-edit screen over four tasks and five paired trials per task. Error bars are 95% Wilson intervals. The preregistered verdict is no-go because retention misses the effect-size gate and removal has the wrong sign.

359 A.8 Numerical scope of rational conversion

360 The deployed Padé computation is exactly rational even though it approximates RMS normalization.
 361 Final certification must report its polynomial degrees, coefficients, denominator margin, observed
 362 activation range, tail error, action drift, and runtime overhead. Operator membership alone does not
 363 establish practical numerical safety under distribution shift.